

Wavelet-Driven Cross-Domain Consistency for Mixed-Supervised 3D Tumor Segmentation

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Abstract. Accurate 3D tumor segmentation is vital for clinical decision-making, but pixel-wise annotation is expensive. Mixed-supervised learning combines a small set of mask labels with abundant box labels, yet reconstruction-kernel heterogeneity creates large domain gaps. We propose WCC4MS, a jointly optimized lesion-centered framework that models kernel-related frequency shifts via a 3D discrete wavelet transform. The input volume is decomposed into low- and high-frequency components, and the scaled high-frequency coefficients generate synthetic smooth and sharp variants. A dual-branch architecture learns from mask and box annotations with task-specific decoders, while a cross-branch KL-divergence constraint enforces semantic consistency. An auxiliary domain-aware head encourages quality-sensitive features and improves robustness to domain shift. At inference, only the segmentation branch is used on candidate-centered patches; the box branch is used only during training to exploit weak annotations. Experiments across three mask-annotated CT datasets and one box-only dataset show consistent gains in segmentation accuracy and cross-domain generalization compared to prior mixed- and weakly supervised methods.

Keywords: Mixed-supervised segmentation · Cross-domain consistency · 3D wavelet transform

1 Introduction

Tumor segmentation supports clinical decision-making by providing a quantitative characterization of lesions and analyzing their progression. The availability of large training datasets with high-quality masks has been a key factor for

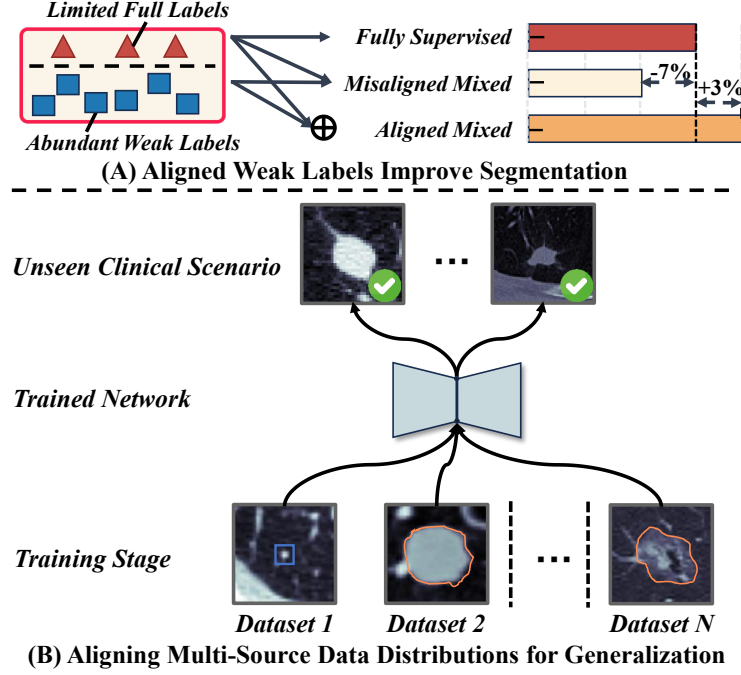


Fig. 1: (A) Appropriately integrating abundant weak annotations into mixed-supervised training improves tumor segmentation efficacy compared to fully supervised approaches. (B) In the novel setting of mixed-supervised learning, supervision signals come from multi-source datasets, making it critical to align their distributions and ensure generalizability to unseen clinical scenarios.

these advances. However, obtaining such annotations is costly, as it requires expert radiologists to perform pixel-wise labeling and cross-validated review to ensure accuracy in 3D computed tomography (CT) images with high variability. To alleviate the need for a large, labeled segmentation dataset, weakly supervised segmentation with cost-effective labels has attracted considerable attention recently. Nevertheless, despite extensive efforts, weakly supervised models generally underperform their fully supervised counterparts, limiting their practicality in real-world clinical applications. Another promising learning scenario is mixed supervision, where only a small fraction of data is densely annotated, and a large dataset contains weakly supervised images. Combining these datasets to train a unified model framework can maximize the use of available annotations. As shown in Figure 1(A), prior work has demonstrated that balancing the task weights across the two supervision types can improve segmentation performance. These mix-supervised approaches outperform fully supervised learning on small-scale datasets [19, 27, 33]. However, existing methods rarely account for distribution differences across datasets that provide different supervision signals [45]. As shown in Figure 1(B), variations in image quality across different datasets can

lead to a significant performance drop in clinical applications. Moreover, simply integrating existing domain adaptation methods into a mixed supervision framework is non-trivial. Image quality assessment and segmentation represent fundamentally distinct objectives: the former focuses on identifying shifts in data distribution, while the latter prioritizes precise tumor edge classification [4]. This fundamental difference poses challenges to existing mix-supervised frameworks.

In this paper, we present WCC4MS, a simple yet effective cost-minimization framework for 3D tumor segmentation in a mixed-supervised setting. Our dual-branch architecture handles mask and box via annotation-type routing and task-specific branches. The framework focuses on lesion-centered segmentation after candidate localization, which is consistent with common 3D CAD pipelines where suspicious regions are first localized and then refined by a lesion-level segmentation model. CT data exhibit significant domain variations due to reconstruction kernels, with high-noise versions preserving sharper edges than low-noise counterparts. Inspired by multi-kernel CT analysis [16], we apply 3D wavelet transforms [42] to decompose input volumes into low-frequency (LF) and high-frequency (HF) components. To approximate kernel-related frequency shifts without raw metadata, we scale the HF components to generate smooth and sharp variants. To enhance robustness against such heterogeneity, a consistency constraint is imposed between the two branches. Furthermore, we introduce a domain classification branch to explicitly predict the data provenance, guiding the encoder to learn quality-aware features and thereby improving cross-domain generalizability and structural consistency.

It is worth emphasizing that our method differs fundamentally from the foundation models [12, 21]. While foundation models aim for universal diagnostic capability and rely on massive data and computational resources across multiple anatomical regions, our approach focuses on maximizing the use of available supervision within limited datasets and computational budgets, making it more practical for clinical deployment. Our contributions are highlighted as follows:

- We propose WCC4MS, a novel mixed-supervised framework for 3D tumor segmentation that unifies mask-supervised and box-supervised learning in a dual-branch architecture.
- We introduce wavelet transforms to model domain shifts and effectively bridge image quality assessment with tumor segmentation tasks to enhance the cross-domain adaptability of the model.
- We conduct extensive experiments on three typical tumor segmentation datasets with one auxiliary box dataset. Our method outperforms state-of-the-art approaches.

2 Related Work

2.1 Medical Mixed-supervised Segmentation

Though stronger labels lead to better segmentations, practical clinical deployment necessitates label-efficient solutions. Traditional approaches often treat different supervision types independently. In contrast, mixed-supervised methods

can leverage massive, low-cost, and weak labels, as well as a few dense labels, to ensure segmentation accuracy. Specifically, recent works address this by decoupling supervision branches, where a basic feature extractor is shared [17, 27]. However, treating these independent branches equally may introduce a high risk of overfitting towards weakly supervised data. Recent semi-supervised medical segmentation methods also explore teacher-student interactions to improve label efficiency, such as DualFete [39], which revisits feedback between teacher and student models. Different from these semi-supervised settings, our work focuses on mixed supervision with heterogeneous mask and box annotations collected from different domains.

2.2 Domain Generalization in Medical Imaging

Different supervision signals typically originate from distinct data sources, and their simultaneous use may introduce domain shift issues. Domain generalization (DG) aims to learn a model from multiple source domains that can directly generalize to unseen target domains [36, 44]. Compared to domain adaptation, DG is better suited for medical imaging problems due to the low interoperability and scarcity of sample diversity characteristic of medical image data [10]. Several approaches focus on learning domain-invariant features by reducing distribution differences between source domains [6, 26, 38]. Alternative meta-learning strategies simulate domain shifts through episodic training, where models are alternately exposed to meta-train and meta-test splits during optimization [9, 13]. Current mixed-supervised models overlook the effects of domain shift on generalization. Our wavelet-based framework decomposes CT images into multi-frequency components, explicitly modeling acquisition variations to improve noise robustness.

2.3 Multi-Task Learning

Multi-task learning has proven to be a practical approach for training generalized models that can address multiple objectives simultaneously. Such joint optimization across spatial and scale domains has shown benefits in diverse vision tasks [41]. In the medical domain, it has been widely adopted to boost diagnostic performance by jointly learning related tasks, such as capturing inter-disease relationships and leveraging auxiliary information [18, 29]. Beyond medical imaging, structure-enhanced correlation modeling has also benefited other multi-task scenarios [15]. Despite its success, a critical aspect often remains underexplored: the role of image quality in medical diagnosis. While some prior work has attempted to include quality assessment as an auxiliary task, they reported limited improvements due to the lack of an explicit mechanism to leverage quality information effectively [27]. In contrast, WCC4MS introduces a framework that explicitly integrates domain knowledge into the diagnostic process, unlocking its potential to enhance model accuracy and robustness.

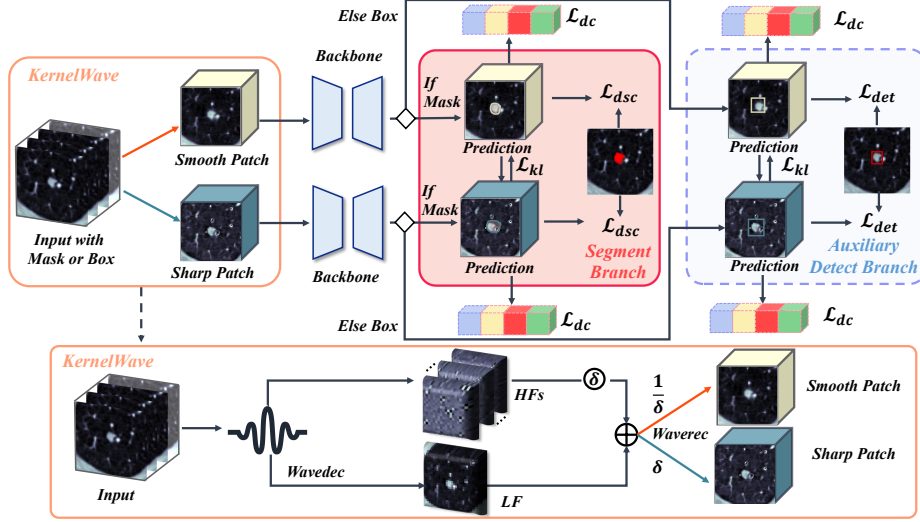


Fig. 2: This figure illustrates the overall WCC4MS framework, where KernelWave generates smooth and sharp patches to mimic kernel variations. A dual-branch architecture jointly learns from mask- and box-supervised data, with consistency and domain-aware losses enhancing cross-domain generalization.

3 Method

3.1 Overview

We focus on maximizing the utilization of multi-datasets with diverse label categories by leveraging the advantages of a dual-branch network to enhance generalization against domain shift. Unlike existing implicitly auxiliary learning [4], we leverage the deterministic information provided by bounding boxes, specifically their positions and sizes, to explicitly enhance the sensitivity of the network to segmentation regions. Figure 2 shows the architecture of the WCC4MS, a framework specifically designed for domain shift in the mixed supervised paradigm.

3.2 KernelWave

The reconstruction kernel is a crucial parameter contributing to domain shifts in CT imaging [16]. It operates as frequency-domain filtering, mathematically characterized via the Fourier transform as the selective amplification or suppression of specific frequency components within CT images. However, due to patient privacy concerns, public datasets typically do not retain raw metadata, making it impossible to unify CT data distributions through the inverse Fourier transform. As a widely employed multiscale frequency-domain tool, wavelet transformation enables decomposition of CT images acquired with unknown kernels into distinct subbands. Adjusting high-frequency subband coefficients enables the generation

of smooth or sharp variants that approximate kernel-related frequency shifts while preserving the original anatomical structure.

Given a 3D medical patch $\mathcal{X} \in \mathbb{R}^{D \times H \times W}$ (D , H , and W indicate the depth, height, and width, respectively), the KernelWave processes the image that undergoes multi-level wavelet decomposition (MWD). At each level $v \in \{1, 2, \dots, V\}$, the decomposition produces one LF approximation subband $\mathcal{X}_{LLL}^{(v)}$ and seven HF detail subbands $\mathcal{X}_\epsilon^{(j)}$, where $\epsilon \in \{LLH, LHL, HLL, LHH, HLLH, HHL, HHH\}$. Formally, the decomposition can be expressed as:

$$\mathcal{X} \xrightarrow{\text{MWD}} \left\{ \mathcal{X}_{LLL}^{(V)}, \left\{ \mathcal{X}_\epsilon^{(v)} \right\}_{\substack{\epsilon \in \{LLH, LHL, \dots, HHH\} \\ v=1, \dots, V}} \right\}. \quad (1)$$

Specifically, the LF subband is:

$$\mathcal{X}_{LLL}^{(V)} = (\mathcal{X} * (\phi_L \otimes \phi_L \otimes \phi_L)) \downarrow 2, \quad (2)$$

then one of the HF subbands $\mathcal{X}_{LLH}^{(j)}$ can be denoted as:

$$\mathcal{X}_{LLH}^{(v)} = (\mathcal{X} * (\phi_L \otimes \phi_L \otimes \psi_H)) \downarrow 2, \quad (3)$$

where ϕ_L is the low-pass scaling function, ψ_H is the high-pass wavelet function, $\otimes$ denotes the tensor product, and $\downarrow 2$ represents downsampling by a factor of 2 along each dimension.

To generate kernel-related smooth and sharp variants, the HF detail coefficients are adjusted by a scaling factor $\delta > 1$, while the LF coefficient remains unchanged. For a sharp kernel image, the modified HF coefficients $\tilde{\mathcal{X}}_\epsilon^{(v)}$ can be obtained by:

$$\tilde{\mathcal{X}}_\epsilon^{(v)} = \delta \cdot \mathcal{X}_\epsilon^{(v)}, \quad (4)$$

the smooth counterpart can be denoted as:

$$\tilde{\mathcal{X}}_\epsilon^{(v)} = \frac{1}{\delta} \cdot \mathcal{X}_\epsilon^{(v)}. \quad (5)$$

Given the modified coefficients, the multi-level wavelet reconstruction (MWR) reconstructs the modified image as follows:

$$\tilde{\mathcal{X}} = \text{MWR} \left(\left\{ \mathcal{X}_{LLL}^{(V)}, \tilde{\mathcal{X}}_\epsilon^{(v)} \right\} \right) \in \mathbb{R}^{D \times H \times W}. \quad (6)$$

Due to potential boundary effects and padding during wavelet processing, the reconstructed volume dimensions may differ slightly from the original dimensions. Thus, the final result can be adjusted through cropping.

3.3 Dual-Branch Mixed Supervision

To effectively leverage heterogeneous supervision while enhancing the adaptability to distribution shifts across multiple data sources, we propose a Dual-Branch

Mixed Supervision (DMS) module. Compared with existing approaches, our design offers two key advantages: first, it enables the model to fully exploit all available deterministic annotations, including both pixel-level masks and bounding boxes, to improve segmentation accuracy; second, by enforcing consistency between the two task-specific branches and explicitly incorporating a domain classification objective, the DMS module enhances the ability to perceive and adapt to kernel variations introduced by KernelWave module. The two branches share the identical backbones but perform task-specific decoding and supervision. Specifically, each segmentation branch is trained with Dice loss $\mathcal{L}_{dsc}$ using full-resolution masks:

$$\mathcal{L}_{dsc} = 1 - \frac{2 \sum_i p_i g_i + \sigma}{\sum_i p_i^2 + \sum_i g_i^2 + \sigma}, \quad (7)$$

where p_i and g_i denote the predicted probability and the ground truth label at voxel i , respectively. σ is a small constant to avoid division by zero. Then, the auxiliary detection branch leverages box annotations via a detection loss $\mathcal{L}_{det} = \mathcal{L}_{cls} + \mathcal{L}_{reg}$ [31]:

$$\begin{cases} \mathcal{L}_{cls} = -\frac{1}{N} \sum_{j=1}^N (1 - p_j) \log(p_j) \\ \mathcal{L}_{reg} = \frac{1}{N} \sum_{j=1}^N \left(1 - \text{IoU}(\hat{b}_j, b_j)\right) \end{cases}, \quad (8)$$

where N represents the number of detected objects, p_j means the predicted class probability for box j . $\hat{b}_j$ and b_j denote the predicted bounding box coordinates and ground truth box, respectively. $\text{IoU}(\cdot, \cdot)$ denotes the Intersection over Union. To enhance consistency between the two predictions, we further impose a KL-divergence loss $\mathcal{L}_{kl}$ across the output distributions for either segmentation branches or detection branches:

$$\mathcal{L}_{kl} = \frac{1}{|\Omega|} \sum_{i \in \Omega} \left[p_i^{(1)} \log \left(\frac{p_i^{(1)}}{p_i^{(2)} + \sigma} \right) + (1 - p_i^{(1)}) \log \left(\frac{1 - p_i^{(1)}}{1 - p_i^{(2)} + \sigma} \right) \right], \quad (9)$$

where Ω is the set of voxels in the predicted 3D volume, $p_i^{(1)}$ and $p_i^{(2)}$ are the foreground probability predicted by the first branch and the second branch at voxel i . σ is still a small constant to avoid numerical instability.

To further enhance the awareness of domain-specific characteristics, we introduce a domain classification loss $\mathcal{L}_{dc}$ by supervising a lightweight domain prediction head for each branch:

$$\mathcal{L}_{dc} = - \sum_{c=1}^C y_k^{(c)} \log \hat{y}_k^{(c)}, \quad (10)$$

where C denotes the total number of datasets, $y_k^{(c)}$ denotes the ground truth one-hot indicator for domain c on sample k . $\hat{y}_k^{(c)}$ illustrates the predicted softmax probability that sample k belongs to domain c . The auxiliary domain head is used as a domain-aware regularizer rather than an adversarial domain-invariant module. It encourages the encoder to retain image-quality and domain-related cues, while the segmentation, detection, and KL losses constrain these cues to remain task-relevant. This design improves robustness without explicitly forcing domain-invariant features. All loss components are jointly optimized:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{dsc}} + \lambda_2 \mathcal{L}_{\text{det}} + \lambda_3 \mathcal{L}_{\text{kl}} + \lambda_4 \mathcal{L}_{\text{dc}}. \quad (11)$$

It is worth noting that, except for the KL divergence terms, all other losses are computed as the summation of contributions from the dual-branch.

4 Experiments

4.1 Datasets

In our experiments, we split each dataset with masks into training and test sets with a 4:1 ratio. The PN9, which contains only box annotations, is used exclusively to provide auxiliary training and is not involved in evaluation.

- **LIDC-IDRI:** The LIDC-IDRI [2] dataset comprises lung cancer screening thoracic CT scans with marked-up annotated tumors. Following the previous work [29], we include only CT scans that meet the LUNA16 selection criteria, resulting in 586 CT scans with a total of 1,131 nodules.
- **MSD-Lung:** The MSD-Lung is part of the MSD [1] collection. It contains 3D CT scans from 63 patients.
- **LNDb:** LNDb is a dataset comprising 294 lung CT scans [25]. The database comprises 1429 unique findings by five radiologists.
- **PN9:** The CT scans in PN9 are obtained by a series of CT manufacturers and corresponding models, which are multidomain [23]. It contains 8798 CT scans with 40439 bounding boxes for nodules.

We focus on lung tumor segmentation for two key reasons: (1) The abundance of publicly available lung CT datasets enables robust cross-validation and statistical power; (2) Lung tumors exhibit high heterogeneity in size (3–95mm), morphology (spiculated/lobulated), and density (GGN/solid), making them a representative benchmark for mixed-supervision learning in medical imaging.

4.2 Implementation and Training Details

The networks are implemented using PyTorch 2.1.1, and the experiments are performed on two Nvidia GeForce RTX 4090 GPUs with 24 GB of memory. We use a stochastic gradient descent (SGD) optimizer with a base learning rate of 0.01, momentum of 0.9, and weight decay of 1e-4. The learning rate is adjusted

Table 1: Performance comparison with other state-of-the-art methods using 3D UNet as the backbone. We evaluate the performance using two metrics: DSC (%) and HD95. The strong-weak annotation ratio during training is 1:2. Bold numbers indicate the best results, while underlined numbers indicate the second-best results.

Type	Methods	LIDC-IDRI		MSD-Lung		LNDb	
		DSC↑	HD95↓	DSC↑	HD95↓	DSC↑	HD95↓
Fully Supervised	3D UNet [28]	77.26	1.92	56.46	<u>7.31</u>	63.24	8.59
	SANPNet [7]	65.31	2.86	46.83	14.46	57.76	14.20
Weakly Supervised	MonoBox [14]	68.03	2.75	47.21	15.56	60.25	11.46
	ReliableMD [8]	68.54	2.73	46.14	14.19	61.21	12.32
	BoxInst [32]	66.86	2.78	48.95	14.93	60.47	11.85
Mixed Supervised	MIST [19]	77.45	1.90	56.18	8.14	<u>64.00</u>	<u>8.27</u>
	ConfKD [17]	73.85	2.47	52.02	12.26	63.18	9.87
	MixSegNet [37]	<u>78.19</u>	1.75	55.36	8.13	63.72	8.39
	WSL4MIS [20]	77.62	<u>1.73</u>	<u>57.01</u>	7.49	63.99	8.63
	MSDN [33]	72.44	2.53	51.94	12.13	62.25	9.76
	Ours	80.32	1.52	57.93	7.06	65.84	7.98

using a cosine annealing scheduler. We use a Haar wavelet with decomposition level $V = 2$. WCC4MS is trained with a batch size of 8 for 80 epochs. Following previous work [17, 27, 40], we assume that the first-stage detection task has already been completed for all annotated nodules, thus not affecting practical clinical applicability. We crop a $96 \times 96 \times 96$ patch centered on the annotation from the entire CT image for both training and testing. During testing, we only use the model with the segmentation branch for inference. Following previous works [38, 45], we evaluate segmentation performance using two standard metrics: the Dice Similarity Coefficient (DSC) in percentages and the 95% Hausdorff Distance (HD95).

4.3 Comparison With State-of-the-Art Methods

Quantitative Results with Label-Efficient Methods We set the strong-weak ratio at 1:2 during the training stage. The PN9 dataset provides weak labels. We compare our approach with five other mixed-supervision methods and four weakly supervised methods designed for medical images. For a fair comparison, all competing methods are evaluated under the same box or mixed-supervision setting with the widely adopted 3D UNet [28] as the backbone and training pipeline. As illustrated in Table 1, the WCC4MS outperforms other methods on all three datasets. Remarkably, it demonstrates superior performance on the LIDC-IDRI dataset, achieving substantial improvements of 2.1% and 3.1% in DSC over the second-best method and the fully supervised method, respectively. Quantitative analysis reveals that WCC4MS more effectively leverages weak annotations compared to existing mixed-supervised and weakly-supervised approaches.

Table 2: We replace backbone 3D UNet by 3D UNetr, while maintaining other setting.

Type	Methods	LIDC-IDRI		MSD-Lung		LNDb	
		DSC $\uparrow$	HD95 $\downarrow$	DSC $\uparrow$	HD95 $\downarrow$	DSC $\uparrow$	HD95 $\downarrow$
Fully Supervised	3D UNetr [11]	79.33	1.80	58.05	8.47	64.89	8.24
	SANPNet [7]	67.19	2.74	50.13	14.13	55.61	13.85
Weakly Supervised	MonoBox [14]	70.11	2.42	49.86	13.14	60.17	11.76
	ReliableMD [8]	71.34	2.56	49.14	13.88	63.27	9.62
	BoxInst [32]	67.43	2.46	50.76	12.64	60.89	10.73
Mixed Supervised	MIST [19]	79.66	1.86	57.41	8.13	64.97	8.13
	ConfKD [17]	74.75	2.29	54.49	11.31	63.61	9.72
	MixSegNet [37]	79.19	<u>1.64</u>	<u>58.36</u>	8.67	<u>65.77</u>	<u>7.86</u>
	WSL4MIS [20]	<u>80.15</u>	1.74	57.39	8.39	64.99	7.98
	MSDN [33]	74.96	2.04	53.79	9.65	63.36	9.32
	Ours	81.16	1.43	60.02	<u>8.21</u>	67.90	7.51

To validate the robustness using a different backbone, we replace the backbone from 3D UNet with the also widely used 3D UNetr [11]. The results in Table 2 demonstrate that the proposed WCC4MS outperforms all compared methods in most tasks, except for the HD95 score in the MSD-Lung dataset, where it slightly underperforms compared to MIST (another recently state-of-the-art method for mixed-supervised segmentation).

Quantitative Results with Cross-Domain Methods To validate the adaptability of our method to diverse data distributions, we compare it with other state-of-the-art domain generalization methods. Following the standard evaluation protocol in domain generalization research, we employ the "leave-one-domain-out" strategy: using the test set from one domain as the target domain and using only the training sets from the other domains as the source domains. The experimental results are presented in Table 3. The DeepAll refers to a 3D UNet without any domain generalization method. It serves as a lower bound. The experimental results demonstrate that WCC4MS achieves state-of-the-art performance across all three datasets, outperforming the second-best method (GTVSeg) by significant margins of 1.9% in DSC and 2.0 in HD95 on average.

Comparison with Segmentation Anything Models For completeness, we additionally compare our method with the Segment Anything Model (SAM) series in Table 4. The results show that our method consistently achieves better lesion segmentation performance, especially for small tumors where SAM-based models often struggle.

Cross-Modality Validation on MRI The MRI experiment is used as a cross-modality extension to test whether the frequency-domain consistency principle generalizes beyond CT. We construct a four-dataset MRI setting using three

Table 3: Performance comparison with other domain generalization methods using 3D UNet as the backbone.

Methods	LIDC-IDRI		MSD-Lung		LNDb		Average	
	DSC↑	HD95↓	DSC↑	HD95↓	DSC↑	HD95↓	DSC↑	HD95↓
DeepAll	59.47	8.49	44.24	18.76	51.27	15.90	51.66	14.38
MKCNet [4]	63.79	6.25	48.63	15.48	54.50	12.18	55.64	11.30
XNet [45]	61.50	6.83	<u>52.71</u>	14.22	<u>56.94</u>	13.61	57.05	11.55
M2CAN [43]	63.44	7.25	51.08	13.84	56.15	<u>11.97</u>	56.89	11.02
GTVSeg [35]	<u>66.53</u>	<u>5.89</u>	52.48	<u>12.76</u>	54.79	13.43	<u>57.93</u>	<u>10.69</u>
Ours	67.21	5.38	53.61	12.40	58.84	11.26	59.88	9.68

Table 4: Performance Comparison with SAM series.

Methods	LIDC-IDRI		MSD-Lung		LNDb	
	DSC↑	HD95↓	DSC↑	HD95↓	DSC↑	HD95↓
MedSAM2 [22]	74.58	2.81	<u>53.97</u>	9.50	<u>63.97</u>	<u>8.12</u>
Ma-sam [5]	75.06	4.19	48.33	10.96	61.67	9.88
SAM-Med3D [34]	<u>78.37</u>	<u>1.99</u>	50.15	<u>8.31</u>	59.14	9.94
Ours	80.32	1.52	57.93	7.06	65.84	7.98

mask-labeled benchmarks (BraTS [3], MSD-BT [1], FeTS [24]) and one box-only set (TCGA-GBM [30]) for training. As summarized in Table 5, the primary CT-kernel motivation remains evaluated on CT datasets, while the MRI results indicate that the same design can also benefit broader protocol-induced appearance shifts.

Qualitative Results To further demonstrate the superiority of our method, we present the qualitative comparison with other state-of-the-art methods in Figure 3. The CT images are acquired from multiple domains, and the nodules present diverse radiological characteristics. However, the proposed WCC4MS accurately identifies subtle boundaries while maintaining consistency with the complex shape.

4.4 Ablation Study

Experiments under Different Strong-Weak Ratio We conduct comparative experiments on the LIDC-IDRI dataset using the DSC as the evaluation metric, comparing existing mixed-supervision methods. With a fixed number of mask-annotated samples, introducing weakly-labeled box data (adjusting the mask-to-box ratio from 1:0 to 1:2) led to performance gains for our method and three other representative approaches. This indicates that a moderate number

Table 5: MRI mixed-supervised segmentation comparison.

Type	Methods	BraTS		MSD-BT		FeTS	
		DSC $\uparrow$	HD95 $\downarrow$	DSC $\uparrow$	HD95 $\downarrow$	DSC $\uparrow$	HD95 $\downarrow$
Fully Supervised	3D UNet [28]	83.26	9.15	80.97	4.46	82.41	8.62
	SANPNet [7]	70.39	12.97	72.53	7.60	68.46	11.61
Weakly Supervised	MonoBox [14]	72.41	12.06	73.54	7.09	69.84	11.87
	ReliableMD [8]	74.59	11.23	73.78	6.88	71.29	10.56
	BoxInst [32]	72.86	11.63	75.90	7.45	73.96	11.85
	MIST [19]	83.52	<u>9.01</u>	80.79	<u>4.14</u>	<u>82.07</u>	<u>8.39</u>
Mixed Supervised	ConfKD [17]	80.56	10.89	79.10	6.24	81.33	9.98
	MixSegNet [37]	79.87	9.98	80.62	5.20	80.90	10.24
	WSL4MIS [20]	82.74	9.53	<u>81.08</u>	4.32	81.73	9.79
	MSDN [33]	<u>83.71</u>	9.37	78.26	5.68	80.55	9.10
	Ours	85.69	8.75	83.04	3.73	83.62	7.78

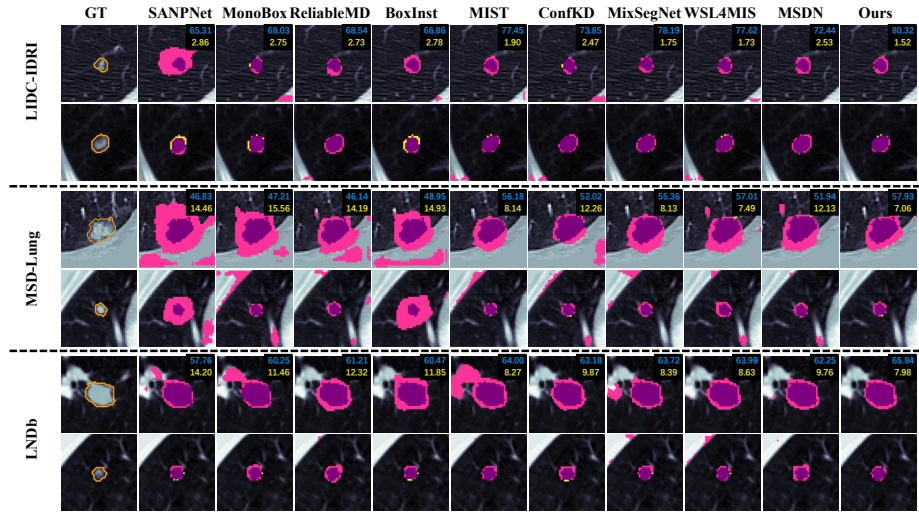


Fig. 3: Visualization results of different methods using 3D UNet as the backbone on three mask datasets. The blue and yellow numbers represent the DSC (%) and HD95 values for the corresponding methods, respectively. True Positives (TP), false positives (FP), and false negatives (FN) are marked by purple, pink, and yellow, respectively. Our method yields smaller FP and FN regions than other methods.

of box-level annotations can effectively enhance segmentation quality. However, as the ratio increased further from 1:2 to 1:4, the DSCs begin a more rapid drop observed near the 1:5 ratio. This trend suggests that excessive box supervision may divert models' attention from fine-grained boundary textures, ultimately degrading segmentation performance.

Table 6: Ablation studies on the proposed components on the LIDC-IDRI dataset. The p -value denotes a statistical significance test for DSC (vs. proposed WCC4MS).

No.	KW	KL	DC	AD	DSC $\uparrow$	HD95 $\downarrow$	p -value
1					63.17	3.74	0.009
2				✓	72.95	2.39	0.015
3	✓	✓	✓		66.27	3.10	0.001
4	✓	✓		✓	76.89	2.03	0.005
5	✓		✓	✓	78.05	1.95	0.036
6		✓	✓	✓	75.46	2.17	0.018
7	✓	✓	✓	✓	80.32	1.52	-

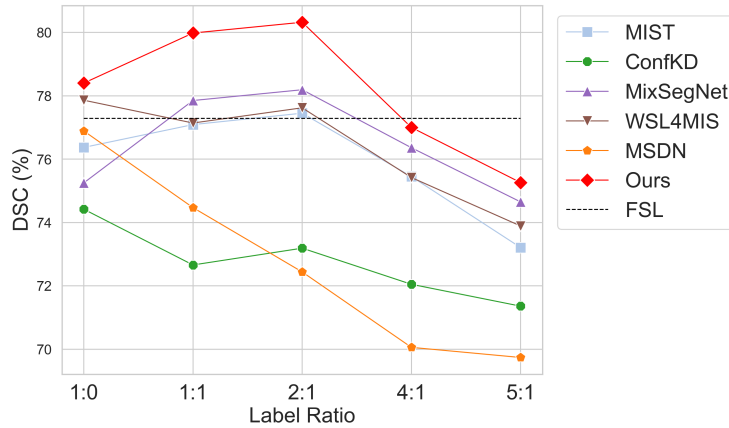


Fig. 4: DSC (%) of methods under the five ratios (box: mask) on the LIDC-IDRI dataset. The black-dashed line represents performance under full supervision.

Contribution of the Key Components We conduct ablation experiments to validate the effectiveness of the four key components: KernelWave (KW), KL consistency (KL), the auxiliary domain-aware regularizer (DC), and the auxiliary detection branch (AD). As shown in Table 6, removing AD causes the largest performance drop, highlighting the importance of exploiting box-level weak labels during training. Removing KW also leads to a larger degradation than removing KL or DC, indicating that approximating kernel-related frequency shifts is important for cross-domain adaptability.

Since disabling AD also removes the mechanism that uses box supervision, we additionally perform an AD-fixed controlled check to isolate KW/KL/DC. Starting from AD-only, adding KW provides the largest gain (+2.04% DSC). Adding KL and DC further improves stability and boundary robustness, reducing the five-seed DSC standard deviation from 0.31 to 0.12 and HD95 from 2.36 mm to 1.54 mm. Removing DC decreases cross-domain DSC by 1.6%, supporting its role as a domain-aware regularizer rather than a redundant classifier.

Table 7: Comparison under different wavelet gains δ , loss weights λ_{1-4} , and frequency augmentation strategies.

(a) Wave gain			(b) Loss weights			(c) Frequency		
δ	DSC $\uparrow$	HD95 $\downarrow$	λ_{1-4}	DSC $\uparrow$	HD95 $\downarrow$	Strategy	DSC $\uparrow$	HD95 $\downarrow$
2	79.26	1.62	(1,1,2,0.1)	79.43	1.97	Laplacian	76.98	2.32
3	80.01	1.56	(1,1,1,0.5)	80.19	1.86	Gaussian blur	79.48	2.15
4	80.32	1.52	(1,1,0.1,0.5)	79.65	1.81	Fourier	77.56	1.98
5	79.94	1.59	(1,1,0.5,0.1)	80.32	1.52	KernelWave	80.32	1.52

Table 8: Mechanism-level validation of KernelWave. RKPN is used only to evaluate real multi-kernel frequency behavior, not as an additional segmentation benchmark.

Analysis	Metric	Result
Real B60f/B31f on RKPN	HF-energy ratio	1.44
KernelWave-sharp/original	HF-energy ratio	1.50
Real vs. KernelWave shift	Spectral corr.	$r = 0.71$
Gaussian blur on LIDC-IDRI	DSC (%)	79.48 ± 0.44
KernelWave on LIDC-IDRI	DSC (%)	80.32 ± 0.15

Mechanism-level validation of KernelWave Although KernelWave is motivated by CT reconstruction-kernel variation, it is not intended to exactly simulate the reconstruction process. Instead, it serves as a metadata-free frequency proxy that approximates the dominant kernel-related effect, i.e., high-frequency amplification or suppression. To verify whether this proxy is consistent with real multi-kernel CT data, we analyze RKPN, a paired B31f/B60f lung CT dataset reconstructed with smooth and sharp kernels. Since RKPN provides detection annotations rather than voxel-wise masks, we use it only for mechanism-level frequency validation, not as an additional segmentation benchmark.

As shown in Table 8, KernelWave-induced sharp variants increase high-frequency energy in the same direction as real B31f/B60f reconstruction differences. The high-frequency energy ratio is 1.50 for KernelWave-sharp/original and 1.44 for real B60f/B31f. The spectral correlation between real and KernelWave-induced frequency changes reaches $r = 0.71$, indicating that KernelWave captures kernel-related frequency shifts rather than arbitrary perturbations. In addition, a five-seed comparison on LIDC-IDRI shows that KernelWave significantly outperforms Gaussian blur, further supporting the benefit of wavelet-based high-frequency modulation over generic smoothing.

Hyperparameters Analysis The results in Table 7 reveal that the model benefits from a proper wavelet gain $\delta = 4$, it yields the best balance between edge enhancement and stability. The optimal loss-weight configuration (1, 1, 0.5, 0.1) suggests that moderate weighting of the auxiliary consistency and domain losses helps regularize training without suppressing supervised gradients. Moreover,

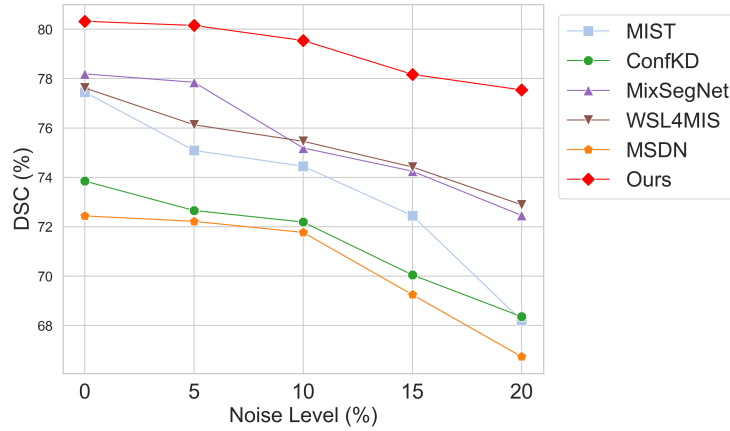


Fig. 5: Robustness against box noise on the LIDC-IDRI dataset.

among various frequency augmentations, KernelWave achieves the best results, confirming the advantage of learning adaptive high-frequency modulation over conventional handcrafted filters.

Robustness Analysis Under Box Perturbations As shown in Figure 5, the noise level denotes the percentage of random perturbation applied to each box’s center position and edge length. This result demonstrates that the WCC4MS effectively suppresses error propagation from noisy weak labels and enhances robustness to annotation inaccuracies. Specifically, under 20% box noise, WCC4MS retains 77.6% DSC while the strongest baseline drops to 72.9%, showing a markedly smaller degradation.

5 Conclusion

We propose a wavelet-driven mixed-supervised framework for cross-domain 3D tumor segmentation. By integrating masks and boxes through a Dual-Branch Supervision module with cross-branch consistency, KernelWave enables domain-aware representation learning that improves robustness to CT reconstruction variations. Experiments across multiple datasets demonstrate superior performance under varied annotation ratios. Future work will extend the framework to other modalities and weaker annotation regimes.

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